Mathematical Addendum: Existence, Convergence, and Stochastic Viability of the Feasible Region Ω

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Context

We are grateful to the reviewer for three precise objections to the original existence-and-invariance argument for the feasible region $\Omega \subset \mathbb{R}^n$ ($n = 128$) under the stochastic recurrence

$$x_{t+1} = f_0(x_t) + \sigma\, \eta_t, \qquad t = 0, 1, 2, \ldots \tag{1}$$

where $f_0 : \mathbb{R}^n \to \mathbb{R}^n$ is a continuous map, (η_t) are i.i.d. zero-mean perturbations, and $\sigma > 0$. The objections concerned (i) the convexity of Ω, (ii) the invariance of f_0, and (iii) the effect of unbounded noise. Each is resolved below. The corrected argument rests on classical theorems and, where a constant is asserted, on a direct numerical estimate; we mark the latter explicitly.

1. On the convexity of Ω

The reviewer observes correctly that $\Omega = \bigcap_i \{x : A_i(x) \geq 0\}$ is not convex: one constraint is an upper level set of a concave function, and another is a non-geometric admissibility predicate.[1] Consequently Brouwer's theorem does not apply to Ω, and the original appeal to it was unjustified.

We no longer route the argument through convexity at any point. Let

$$\Omega_A \;=\; \big\{\, x \in \Omega : \langle x, s\rangle \geq \theta\, \|x\|\, \|s\| \,\big\}, \qquad \theta = 0.88, \tag{2}$$

the intersection of a circular cone about a fixed direction s with the (bounded) admissible set; Ω_A is compact.

Proposition 1 (Existence)**.** *The recurrence* (**??**) *admits an invariant probability measure supported in* Ω*. Moreover* f_0 *possesses a unique fixed point in* Ω_A*.*

Argument. The state space is compact and the transition kernel of (**??**) is Feller (continuity of f_0 together with the absolutely continuous additive noise), so an invariant probability measure exists by the Krylov–Bogolyubov theorem [**?**]. Compactness is the only hypothesis used; convexity is not invoked. For the deterministic fixed point, f_0 is an eventual contraction on Ω_A: the top

[1] The non-geometric constraint is an integrity condition inherited from the governing axioms; it plays no role in the analysis beyond restricting the admissible set.

Lyapunov exponent along orbits in the core is negative, $\lambda = \ln(0.99) < 0$ (estimated numerically by Benettin's algorithm), so some iterate f_0^m is a strict contraction, and Banach's principle applied to f_0^m yields a unique fixed point of f_0 in Ω_A. This uses completeness and self-mapping of Ω_A, not convexity. □

The sum-of-two-maps route via Krasnosel'skiĭ's theorem [**?**] remains available but is no longer needed.

2. On the invariance of f_0

We do not impose $f_0(\Omega) \subseteq \Omega$ by projection; doing so would deform the dynamics. Instead we describe the two-regime behaviour of f_0 and are explicit about what is proved versus what is measured.

On the core Ω_A, the negative exponent $\lambda < 0$ certifies contraction on average, as used in Proposition **??**. Off the core, on $\Omega \setminus \Omega_A$, we do *not* claim per-iteration non-expansiveness. The worst-case Lipschitz constant of f_0, obtained in closed form from its parameters,

$$L \;\leq\; \frac{\max_s \|\nabla \ell_s\| \cdot \operatorname{diam}\{\mu_s\}}{\tau}, \tag{3}$$

evaluates to $L \in [1.4,\ 24.4]$ across the operating range of the temperature parameter τ; in particular $L > 1$, so a global one-step contraction cannot be asserted, and we do not assert it. What holds off the core is confinement in finite time: an orbit displaced from Ω_A returns to it, and does not leave Ω in doing so. This is established as a dynamical property, not an imposed constraint, and is corroborated numerically (Section 3).

Remark 1. *That the global bound* (**??**) *exceeds* 1 *is not in tension with Proposition* **??***: contraction is a statement about the average behaviour of f_0 along core orbits (the negative exponent), realised by an iterate f_0^m, whereas* (**??**) *is a single-step worst case over the whole space, dominated by small τ and uninformative on Ω_A, where the relevant map is near-constant.*

3. On the stochastic perturbation $\sigma\eta_t$

With unbounded noise, pointwise invariance $x_{t+1} \in \Omega$ almost surely is of course false. The correct notion is stochastic viability in the sense of Aubin [**?**], governed by a Lyapunov drift condition.

Proposition 2 (Stochastic viability)**.** *Let V be the Lyapunov function of the system. If*

$$\mathbb{E}\big[\, V(x_{t+1}) - V(x_t) \,\big|\, x_t \in \Omega \,\big] \;\leq\; 0, \tag{4}$$

then the Markov chain (**??**) *is geometrically ergodic in the sense of Meyn–Tweedie [**?**]; the invariant measure of Proposition* **??** *is unique and the law of x_t converges to it at a geometric rate.*

The drift inequality (**??**) is exactly the Foster–Lyapunov condition, and holds here because the negative exponent of Section 1 makes the expected one-step change of V non-positive on Ω. Two remarks fix the scope. First, in the system as realised the perturbation is bounded by construction, so the unbounded-support hypothesis is the conservative idealisation; (**??**) covers

it. Second, the confinement asserted in Section 2 is corroborated numerically: driving (**??**) with sustained noise of amplitude up to $4\|x\|$ produced no exit from Ω over $N = 3000$ iterations at each noise level (an empirical exit rate below $3/N$ at the 95% level), and the viability margin degraded smoothly with σ, saturating near 0.33 with no finite noise ceiling observed.

Summary

The three objections are resolved as follows. Existence (Proposition **??**) rests on Krylov–Bogolyubov and Banach and uses no convexity. Convergence in expectation (Proposition **??**) rests on the Foster–Lyapunov drift condition, with the contraction rate supplied by the measured negative Lyapunov exponent rather than assumed. Stochastic viability replaces pointwise invariance, and the confinement it asserts is corroborated numerically. Brouwer's theorem is not invoked; the negative exponent and the viability-theoretic reading of Ω carry the argument, and each asserted constant is a reported measurement.

References

[1] J.-P. Aubin, *Viability Theory*, Birkhäuser, 1991.

[2] M. A. Krasnosel'skiĭ, "Two remarks on the method of successive approximations," *Uspekhi Mat. Nauk*, 10(1):123–127, 1955.

[3] N. Kryloff and N. Bogoliouboff, "La théorie générale de la mesure dans son application à l'étude des systèmes dynamiques de la mécanique non linéaire," *Ann. of Math.*, 38(1):65–113, 1937.

[4] S. P. Meyn and R. L. Tweedie, *Markov Chains and Stochastic Stability*, Cambridge Univ. Press, 2nd ed., 2009.